

Background Exercises

AE 513: Multivariable Control — Autumn 2025

Linear Algebra

Q1. Let $A = \begin{bmatrix} 4 & 2 \\ 0 & 1 \end{bmatrix}$. What are the eigenvalues of A ?

- (a) $\{4, 2\}$
- (b) $\{6, -1\}$
- (c) $\{1, 1\}$
- (d) $\{4, 1\}$

Q2. Let $B = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. Which vector is an eigenvector with eigenvalue 4?

- (a) $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
- (b) $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$
- (c) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
- (d) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Q3. Let $A \in \mathbb{R}^{2 \times 3}$, $B \in \mathbb{R}^{3 \times 2}$, $x \in \mathbb{R}^2$. Which statement is correct?

- (a) $AB \in \mathbb{R}^{3 \times 3}$; $BA \in \mathbb{R}^{2 \times 2}$; Ax defined; Bx undefined.
- (b) $AB \in \mathbb{R}^{2 \times 2}$; $BA \in \mathbb{R}^{3 \times 3}$; Ax *undefined*; $Bx \in \mathbb{R}^3$.
- (c) $AB \in \mathbb{R}^{2 \times 3}$; $BA \in \mathbb{R}^{3 \times 2}$; both Ax, Bx defined.
- (d) Only Ax is defined.

Q4. Which identity holds for any conformable A, B ?

- (a) $(AB)^\top = A^\top B^\top$
- (b) $(AB)^\top = B^\top A^\top$
- (c) $(AB)^\top = AB$
- (d) None of the above

Q5. Determine the definiteness of the matrix $D = \text{diag}(2, 5)$.

- (a) Positive-Semi-Definite
- (b) Positive-Definite
- (c) Negative-Definite
- (d) Indefinite

Q6. For any real M , which statements are always true?

- (a) $M + M^\top$ is symmetric
- (b) $M - M^\top$ is symmetric
- (c) $(M + M^\top)^\top = M + M^\top$
- (d) Both (b) and (c)

Q7. Let T be invertible and $C = TAT^{-1}$. Which are always equal for A and C ?

- (a) trace and determinant
- (b) trace only
- (c) determinant only
- (d) all entries

Q8. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$. What is $\text{rank}(A)$?

- (a) 0
- (b) 1
- (c) 2
- (d) 3

Q9. Which matrix is orthogonal?

- (a) $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$
- (b) $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$
- (c) $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$
- (d) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Q10. What is D^{-1} for $D = \text{diag}(2, 5)$?

- (a) $\text{diag}(2, 5)$
- (b) $\text{diag}(5, 2)$
- (c) $\text{diag}\left(\frac{1}{5}, \frac{1}{2}\right)$
- (d) $\text{diag}\left(\frac{1}{2}, \frac{1}{5}\right)$

Multivariable Calculus

Q11. For $f(x, y) = x^2y$, compute $\nabla f(x, y)$.

- (a) $(x^2, 2xy)$
- (b) $(2xy, x^2)$
- (c) $(2x, y)$
- (d) $(2x, 2y)$

Q12. For $f(x, y) = e^xy^2$, compute $\nabla f(x, y)$.

- (a) $(y^2e^x, 2ye^x)$
- (b) $(e^x, 2y)$
- (c) $(y^2, 2y)$
- (d) $(y^2, 2ye^x)$

Q13. For $F(x, y) = (x^2, xy)$, find the Jacobian $J_F(x, y)$.

- (a) $\begin{bmatrix} 2x & 0 \\ y & x \end{bmatrix}$
- (b) $\begin{bmatrix} 2x & y \\ 0 & x \end{bmatrix}$
- (c) $\begin{bmatrix} x & 0 \\ y & 2x \end{bmatrix}$
- (d) $\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$

Q14. For $G(x, y, z) = (x + y, x + z)$, compute $J_G(x, y, z)$.

- | | |
|--|--|
| (a) $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ | (b) $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ |
| (c) $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ | (d) $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ |

Q15. For $f(x, y) = 3x^2 + 2xy + y^2$, compute the Hessian $Hf(x, y)$.

- (a) $\begin{bmatrix} 6 & 2 \\ 2 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 6 & 1 \\ 2 & 2 \end{bmatrix}$
(c) $\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 6 & 2 \\ 1 & 2 \end{bmatrix}$

Q16. For $f(x, y) = x^3 + y^2$, compute the Hessian $Hf(x, y)$.

- (a) $\begin{bmatrix} 6x & 0 \\ 0 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}$
(c) $\begin{bmatrix} 3x^2 & 0 \\ 0 & 2y \end{bmatrix}$ (d) $\begin{bmatrix} 6x & 3x^2 \\ 0 & 2 \end{bmatrix}$

Q17. For $f(x, y) = x^2 + y^2$, find the critical point(s) and classify.

- (a) No critical points; function increases everywhere
(b) Single critical point at $(0, 0)$; strict local minimum
(c) Single critical point at $(0, 0)$; strict local maximum
(d) Single critical point at $(0, 0)$; saddle point

Q18. For $f(x, y) = xe^y$, compute $\frac{\partial f}{\partial x}(1, 0)$.

- (a) 0
(b) 1
(c) e
(d) e^1

Q19. Let $x = r \cos \theta$, $y = r \sin \theta$. For $f(x, y) = x^2 + y^2$, compute $\frac{\partial}{\partial r} f(x(r, \theta), y(r, \theta))$.

- (a) 0
(b) r
(c) 2
(d) $2r$

Q20. For the map $(u, v) \mapsto (x, y) = (2u, 3v)$, compute $\det(\partial(x, y)/\partial(u, v))$.

- (a) 0
(b) 1
(c) 6
(d) -6

Optimization

Q21. Minimize $f(x) = x^2 + 2x + 1$ over $x \in \mathbb{R}$. What is the minimizer x^* ?

- (a) -1
- (b) 0
- (c) 1
- (d) No minimizer

Q22. Minimize $f(x, y) = (x - 1)^2 + (y + 2)^2$ over $\mathbb{R}^2$. What is (x^*, y^*) ?

- (a) $(1, -2)$
- (b) $(-1, 2)$
- (c) $(1, 2)$
- (d) $(0, -2)$

Q23. For $f(x, y) = x^2 + y^2 + x$, solve $\nabla f = 0$ and give the minimizer.

- (a) $(\frac{1}{2}, 0)$
- (b) $(-\frac{1}{2}, 0)$
- (c) $(0, 0)$
- (d) $(0, -\frac{1}{2})$

Q24. For $f(x) = ax^2 + bx + c$ with $a > 0$, which statement is true?

- (a) f is concave on $\mathbb{R}$ and has no minimizer
- (b) f is convex on $\mathbb{R}$ and has a unique global minimizer $x^* = -\frac{a}{2b}$
- (c) f is convex on $\mathbb{R}$ and has a unique global minimizer $x^* = -\frac{b}{2a}$
- (d) f is neither convex nor concave

Q25. Minimize $(x - 3)^2$ subject to $x \geq 5$. What is the minimizer x^* ?

- (a) 3
- (b) 4
- (c) 5
- (d) No minimizer

Q26. Which function is convex on $\mathbb{R}$?

- (a) $-x^2$
- (b) e^x
- (c) x^3
- (d) $\sin x$

Basic Python

Q27. What does this print?

```
def add(a, b):  
    print(a + b)  
  
x = add(2, 3)  
print(x)
```

- (a) Prints 5 only
- (b) Prints `None` only
- (c) Prints 5 then `None`
- (d) Raises an error

Q28. Let `nums = [10, 20, 30, 40, 50]`. What is `nums[4:0:-2]`?

- (a) `[50, 30]`
- (b) `[50, 30, 10]`
- (c) `[40, 20]`
- (d) Error

Q29. What does this print?

```
def append_item(x, lst=[]):  
    lst.append(x)  
    return lst  
  
print(append_item(1))  
print(append_item(2))
```

- (a) `[1]` then `[2]`
- (b) `[1]` then `[1, 2]`
- (c) Two separate lists: `[1]` then `[2]`
- (d) Error

Q30. What does this print?

```
m = [[0]*3]*2  
m[0][1] = 9  
print(m)
```

- (a) `[[0, 9, 0], [0, 0, 0]]`
- (b) `[[0, 9, 0], [0, 9, 0]]`
- (c) `[[0, 0, 0], [0, 0, 0]]`
- (d) Error

Q31. What does this print?

```
a = [1, 2, 3]
b = a
a.append(4)
print(b)
```

- (a) [1, 2, 3]
- (b) [1, 2, 3, 4]
- (c) None
- (d) Error

Q32. (Column vector) You have `v = [1, 2, 3]`. Create a NumPy column vector with shape `(3, 1)`. Which snippet does that?

- (a) `np.array(v)`
- (b) `np.reshape(v, (1, 3))`
- (c) `np.array(v).T`
- (d) `np.array(v).reshape(-1, 1)`

Q33. (Matmul shape) If `A.shape == (2, 3)` and `x.shape == (3,)`, what is the shape of `A @ x`?

- (a) `(2,)`
- (b) `(2, 1)`
- (c) `(1, 2)`
- (d) Error